Qp Code: 25901116

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBOT200

ARCHEGONIATES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Maximum Marks: 50

Part A

Short answer

Answer any 10 questions

Each question carries 1 marks

1. Which generation produces spores in archegoniates ?

Ans: Sporophytic generation

2. . Name one type of cell seen in pericycle of *Pinus* needles.

Ans: Albuminous cells, transfusion tissue (Any one -**Liberal valuation**)

3. What is fragmentation?

Ans:A form of asexual reproduction where the plant body breaks into pieces, and each fragment develops into a new individual.

4. What is dictyostele?

Ans: A type of siphonostele where the vascular cylinder is broken into a network of distinct vascular strands (meristeles) by overlapping leaf gaps.

5. What is homospory?

Ans: The production of only one type of spore, which are all equal in size and form.

6. Which pteridophyte is used as a biofertilizer in paddy fields?

Ans: *Azolla*

7. What is peat moss?

Ans: *Sphagnum*, known for accumulating and forming peat. (**Uses of peat moss can also be considered for full mark)**

8. In which year was the modern classification of gymnosperms proposed by Christenhusz et al.?

Ans: 2011

9. State whether female gametophyte is dependent or non-dependent on *Pinus* sporophyte.

Ans: The female gametophyte is dependent on the sporophyte.

10. Which plant group is the first vascular plant group to have appeared on Earth?

Ans: Pteridophytes

11. What is the key characteristic of the gametophyte in Hepaticopsida?

Ans: Prostrate, dorsiventrally flattened , dichotomously branched thallus

12. Name the male and female sex organs in the gametophyte of *Psilotum*

Ans: Antheridia (male) and archegonia (female)

Part B

Short Essay

Answer any 6 questions

Each question carries 5 marks

13. Explain progressive dominance and independence of the sporophyte in archegoniates.

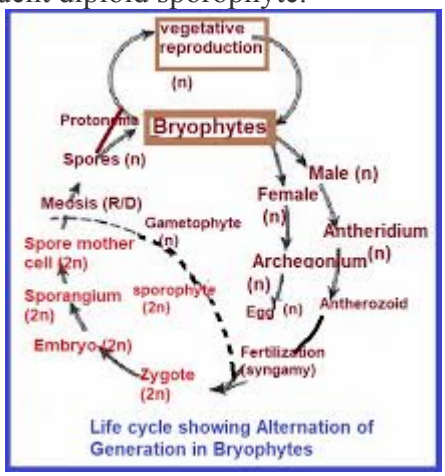
Ans: The progressive dominance and independence of the sporophyte is a fundamental evolutionary trend in archegoniates (bryophytes, pteridophytes, and gymnosperms). This involves a shift from a short-lived, dependent sporophyte in bryophytes to a dominant, free-living sporophyte in vascular plants, allowing for better colonization of land.

Dependent Sporophyte in Bryophytes -In bryophytes (liverworts and mosses), the sporophyte is physically attached to the gametophyte and relies on it entirely or partially for nutrition and water

Transition towards Independence in hornworts- This group shows a crucial intermediate step where the sporophyte has an intercalary meristem (allowing continuous growth) and is photosynthetic**.**

Progressive Dominance & Independence in Pteridophytes- The sporophyte is perennial and showing unlimited growth and completely independent of the gametophyte
Ultimate Dominance in Gymnosperms and Angiosperms- The gametophyte is extremely reduced (e.g., pollen grains and embryo sac) and relies entirely on the sporophyte for survival and the dominant sporophyte is perfectly adapted to dry environments

14. Illustrate the life cycle of bryophytes.
Bryophytes exhibit a heteromorphic alternation of generation featuring a dominant, independent haploid gametophyte and a dependent diploid sporophyte.

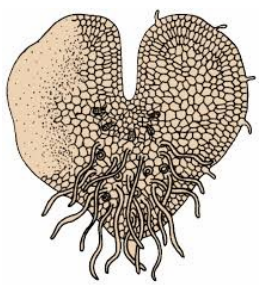


Life cycle showing Alternation of Generation in Bryophytes

15. Give a detailed account of the mature antheridium of Anthoceros
A mature antheridium of Anthoceros is a club-shaped or pouch-like male reproductive organ developed dorsally inside an enclosed, mucilage-filled antheridial chamber. It consists of a multicellular stalk and a main body, enveloped by a single-layered sterile jacket, which changes from green to bright red or orange at maturity, and it contains numerous small, unicellular, biflagellate, curved antherozoids.

16. Give an account on PPG system of classification.
The Pteridophyte Phylogeny Group (PPG) classification (2016), is a modern, community-derived, molecular-based taxonomic system for extant ferns and lycophytes. It organizes these plants based on monophyletic relationships, recognizing two distinct classes: Lycopodiopsida (Lycophytes like club mosses, spike mosses,etc.) and Polypodiopsida (horsetails, ferns,etc), covering 51 families and over 11,900 species. It focuses only on extant (living) plants

17. Describe the structure of *Pteris* prothallus.
Prothallus of *Pteris* is an inconspicuous, small, multicellular heart shaped gametophyte that is independent of the main plant body. It grows in moist, shaded areas, bearing both male (antheridia) and female (archegonia) reproductive organs. It is dorsiventrally flattened, generally one cell thick at the margins, and produces rhizoids on the ventral surface for anchoring and absorption.



(Correct description without diagram can also be considered for full marks)

18. Write short notes on ornamental gymnosperms with examples.

Ornamental gymnosperms are widely used in landscaping and interior design due to their unique foliage, symmetrical growth patterns, and year-round greenery. Unlike angiosperms, they don't produce showy flowers, but they offer architectural beauty

1.**Evergreen Nature:** Most gymnosperms retain their leaves throughout the year, providing consistent color in gardens.

2.Architectural Form: Many species exhibit distinct geometric shapes (conical, pyramidal, or weeping) that serve as focal points.

3. Foliage Diversity: From the needle-like leaves of pines to the fan-shaped leaves of Ginkgo, they offer varied textures.

4. Hardiness: They are often more resistant to harsh weather and poor soil conditions than many flowering plants.

Eg. *Cycas revoluta*, *Thuja* sp., *Cupressus* sp. , *Araucaria* sp., *Juniperus* sp., *Cedrus* sp.,etc.(**Full marks can be given to ornamental values of any five gymnosperms**)

19. Explain the anatomy of capsule in *Pogonatum.*

Anatomical features of the Capsule of *Pogonatum* is as follows-

1)Theca (Main Body): The central part where spores are produced. It consists of

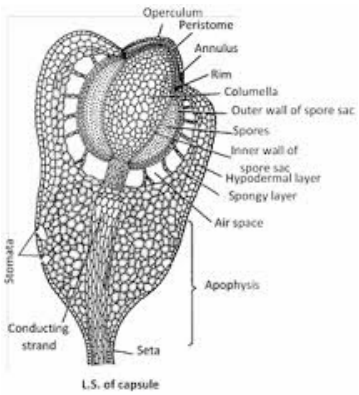
- Wall/Jacket: The capsule wall is 4-5 layers thick. The epidermis is thick-walled, with no stomata present, a key feature in distinguishing Pogonatum from closely related Polytrichaceae.
- Spore Sac: A cylindrical, chambered spore sac exists between the outer wall and the inner columella.
- Columella: A central, sterile column of parenchyma cells that extends from the apophysis to the operculum.

2) Apophysis: The basal, sterile part of the capsule, separated from the main body (theca) by a slight constriction. It connects the seta to the capsule.

3) Operculum and Peristome: The operculum is the lid, often with a rostrate (beaked) apex, covering the capsule. The mouth is surrounded by 32 or sometimes 64 short, triangular, and dark-colored peristome teeth.

4) Epiphragm-A membrane formed by the expansion of the columella, stretching across the mouth beneath the operculum and connecting to the peristome teeth.

5) Calyptra: A dense, hairy covering that protects the young capsule, which is a defining feature of the genus.



L.S. of capsule

(Full marks shall be given for detailed description even without diagram or a brief description with correct diagram)

20. Describe the structure and distinctive features of the megasporophyll of Cycas.

Ans: Megasporophylls are loosely arranged on the crown of the stem, in female Cycas plants , not in a compact strobilus like in male Cycas.The megasporophyll of Cycas is a leaf-like, brownish structure that serves as the female reproductive organ, bearing ovules on its margins . It is differentiated into a leafy, serrated upper lamina, a central ovule-bearing stalk that bears 1-6 pairs (or up to 12) of large, sessile ovules (megasporangia), and a lower petiole, arranged in a crown-like rosette. Unlike male cones, female megasporophylls are loosely arranged in a crown on the crown of the stem, not in a compact strobilus.
The ovules are orthotropous (straight), unitegmic (single integument), and the largest in the plant kingdom.
The entire megasporophyll is typically covered in yellow or brown hairs (tomentum). The megasporophyll is highly variable between species, often with the upper part being narrow and dissected.

Part C
Essay
Answer any 1 questions
Each question carries 10 marks

21. With the help of labelled diagram, explain the structure of strobilus in Selaginella.

The genus Selaginella, is a heterosporous vascular plant that produces two distinct types of spores (microspores and megaspores). These spores are housed within specialized sporangia located at the tips of aerial branches, forming a compact, conical structure known as the strobilus or cone. The strobilus represents a significant step in the evolution of heterospory and the seed habit, concentrating the reproductive organs at the apex for efficient spore dispersal.

1. General Structure and Morphology

The strobilus of Selaginella is a small, terminal, spike-like structure. It is usually found on the tip of lateral branches, though in some species, the stem continues to grow beyond the strobilus, where a fertile zone is interspersed with vegetative leaves. The core axis of the strobilus is a continuation of the stem, often called the strobilar axis, which bears sporophylls in close proximity, creating a compressed, dense appearance compared to the leafy stem.Sporophylls are modified leaves that are typically similar in appearance to ordinary vegetative leaves but are slightly smaller and more pointed. They are generally arranged spirally or in four longitudinal rows along the central axis.
Ligule: A small, membranous structure called a ligule is present on the adaxial surface of every sporophyll at its base, located between the ligule pit and the sporangium.

2. Spore-Bearing Organs (Sporangia)

Each sporophyll bears a single stalked sporangium on its upper (adaxial) surface, typically in the axil.
Microsporangia: These are small, rounded, or oval structures, often reddish or brown, that produce a large number of minute microspores. They are borne on microsporophylls.
Megasporangia: These are generally larger, four-lobed, paler (yellow, greenish-white, or orange) structures that typically produce only four large megaspores. They are borne on megasporophylls.

3. Arrangement within the Strobilus

The arrangement of microsporophylls and megasporophylls inside the strobilus varies by species:
In many species, microsporophylls and megasporophylls are mixed irregularly throughout the strobilus.
Insome species like *S. kraussiana* or *S. rupestris*, the megasporophylls are restricted to the base of the strobilus, while the upper portion consists entirely of microsporophylls.Most Selaginella species have bisporangiate strobili (containing both types). However, in some species the strobili are unisporangiate, meaning a single strobilus produces only one type of spore.

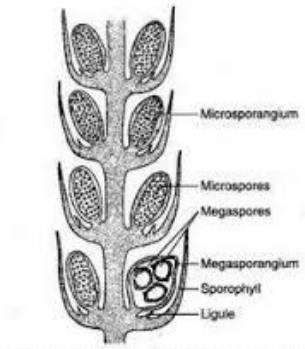


Fig. 7.51 : *Selaginella kraussiana* : L.S. of strobilus

(Description 7 marks+ labelled diagram - 3marks)

22. Describe the life cycle of Cycas, from sporophyte to gametophyte, highlighting reproductive adaptations.

The life cycle of Cycas is haplodiplontic, dominated by a long-lived sporophyte (diploid) generation, while the gametophyte (haploid) is highly reduced and dependent.

1. Sporophyte (Dominant Phase)

The palm like plant body is the diploid sporophyte which is dioecious,

Male Plant -Produces a large cone called a microsporangiate strobilus which contains microsporophylls bearing microsporangia.

Female Plant-Does not form a true cone (unlike many gymnosperms) and bears megasporophylls arranged in a rosette at the apex. Each megasporophyll carries ovules (megasporangia) on its margins.

2. **Formation of Gametophytes**

(A) Male Gametophyte Development

Inside microsporangia, microspore mother cells undergo meiosis → form haploid microspores (pollen grains). Each pollen grain develops into a male gametophyte, consisting of prothallial cells and generative cell which forms two large, motile sperm (antherozoids)

(B) Female Gametophyte Development

Inside the ovule, a megaspore mother cell undergoes meiosis and forms 4 megaspores (only one functional).

The functional megaspore develops into the female gametophyte (haploid endosperm).

pWhich roduces archegonia, each containing a large egg cell.

3. Pollination

Pollination is usually wind-assisted (sometimes insect-aided). Pollen grains reach the ovule through the micropyle. A pollination drop helps to capture pollen.

4. Fertilization (Unique Feature)

The pollen grain germinates, forming a pollen tube (mainly for nutrition, not direct sperm delivery).

The generative cell produces large, multiciliated motile sperms. These sperms swim through a fluid medium inside the ovule to reach the egg. This is a primitive trait, linking Cycas to lower plants.

5. Seed Formation

After fertilization zygote (2n) develops into an embryo and the ovule becomes a seed.

The seed contains: Embryo (young sporophyte) , haploid endosperm (nutritive tissue) and seed coat

6. Germination

The seed germinates to form a new sporophyte plant, completing the cycle.

Reproductive Adaptations of Cycas

- Dioecious condition which promotes cross-pollination.
- Large, protected ovules for better survival of the female gametophyte.
- Retention of megaspore within ovule leads to protection and nourishment.
- Motile sperm and pollen tube marks a combination of primitive and advanced traits.
- Seed formation is the major evolutionary advancement for terrestrial life.
- Reduced gametophyte and dependence on sporophyte increases efficiency.

(Life cycle- 8 marks, Reproductive adaptations (any two)- 2 marks)

****